

Agentic AI

AI That Can Think, Plan and Act

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What is AI?

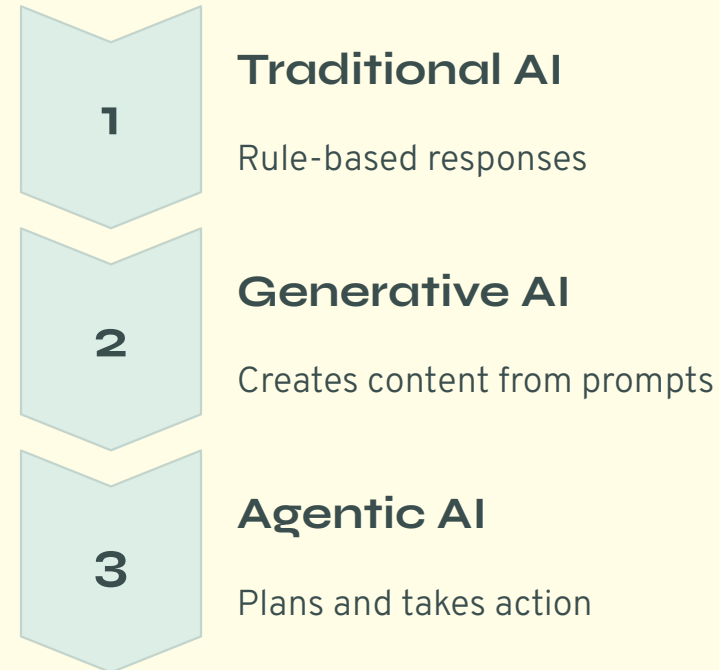
Artificial Intelligence means machines that can **learn and make decisions**.

We already use AI every day:

- Netflix recommendations
- Google Search
- Voice assistants (Siri, Alexa)

Traditional AI mainly **responds to specific inputs** and gives an answer. Now, AI has moved toward systems that can perform **multiple tasks on their own**.

The Evolution



What is Agentic AI?

Agentic AI is an AI system that can **work toward a goal by itself**. It doesn't just answer questions – it plans and takes action.

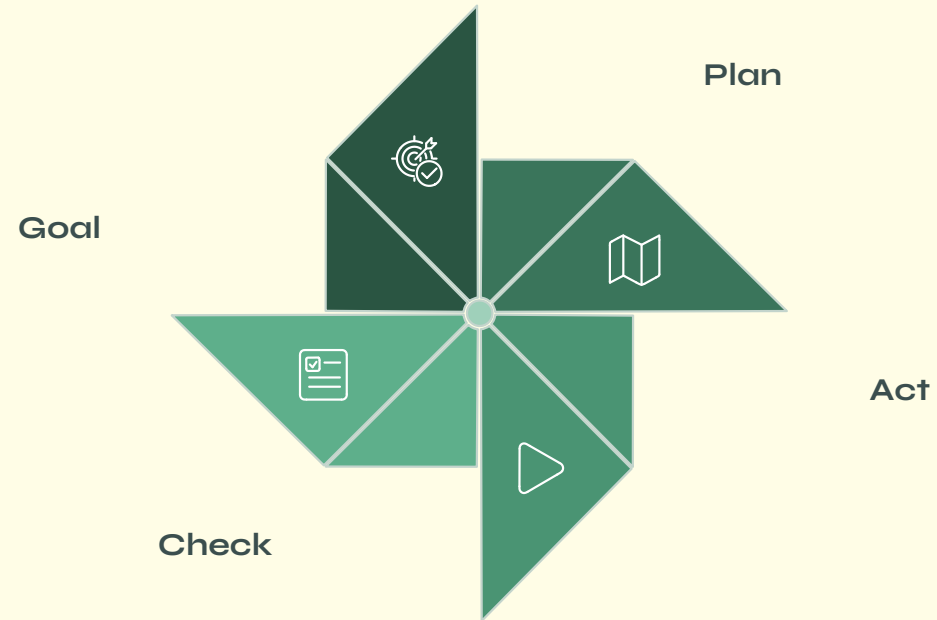
Works toward a goal

Breaks problems into steps

Makes decisions along the way

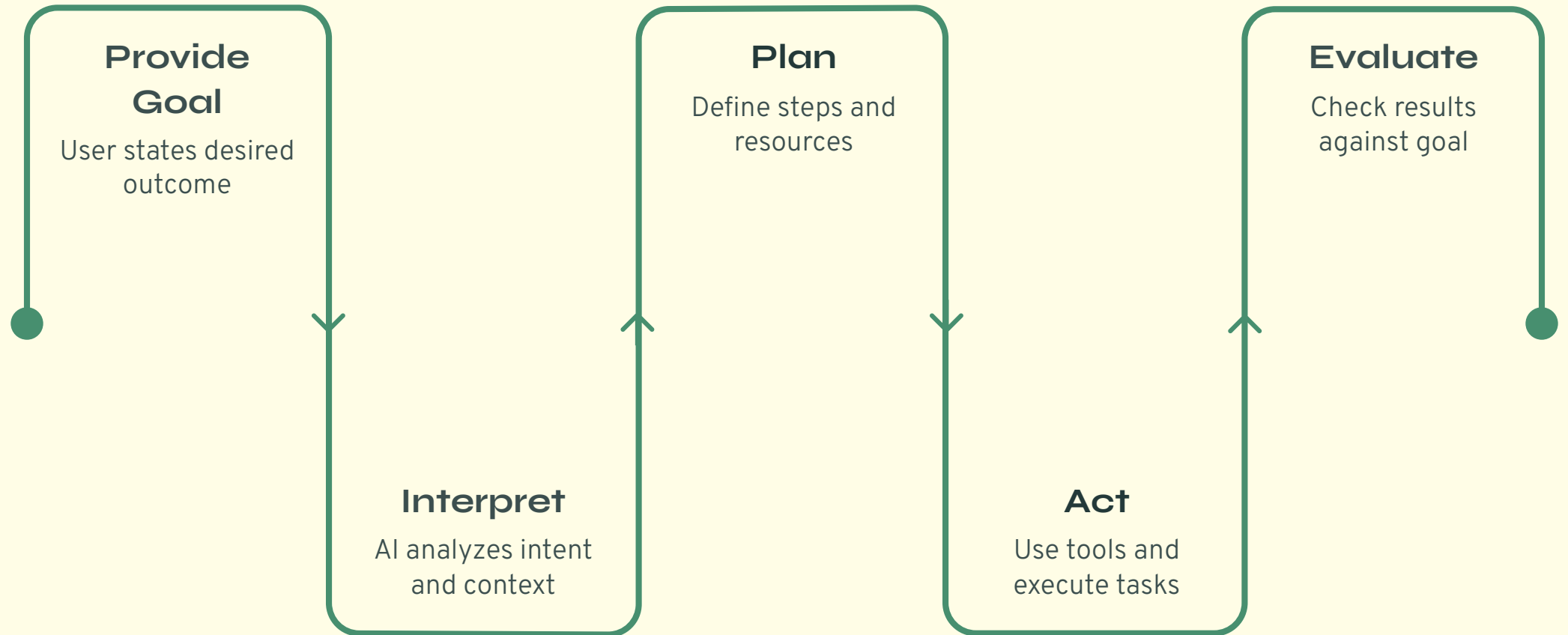
Uses tools and resources

Checks results and improves



How Does Agentic AI Work?

A simple step-by-step process:



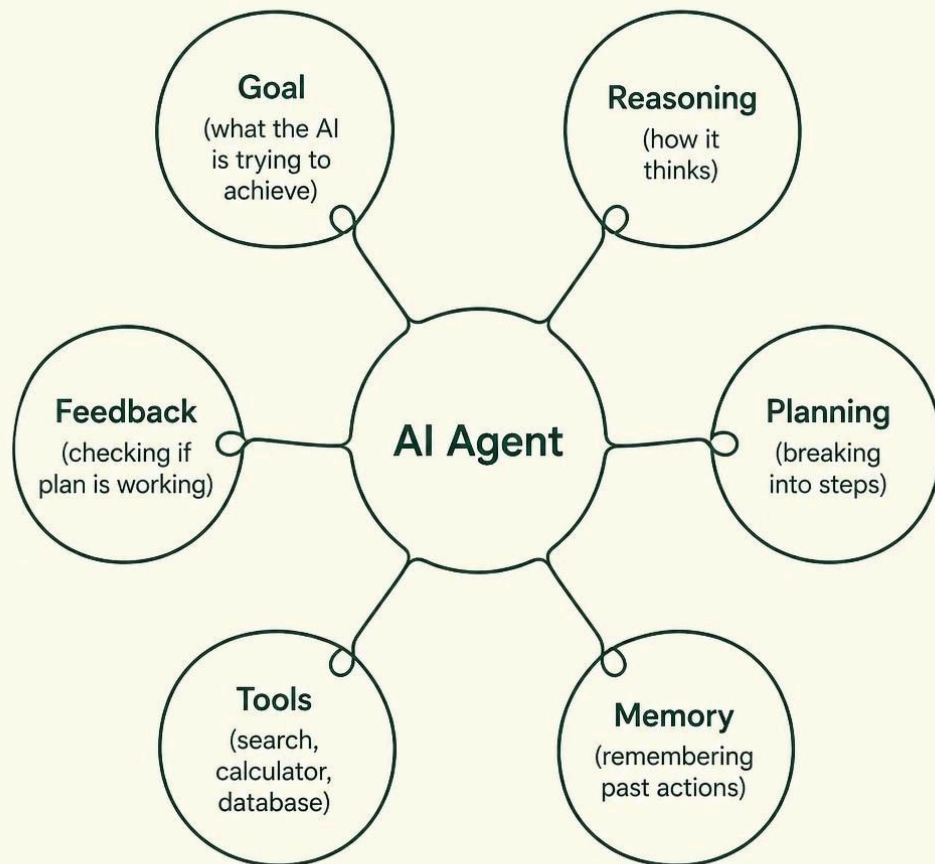
Generative AI vs. Agentic AI

What's the difference?

	Generative AI	Agentic AI
Main Job	Mainly generates content	Works toward a goal
How it Works	Usually responds to prompts	Can perform multiple steps
Output	Produces an answer	Can take real actions
Example	Writing an email	Managing a complete workflow

Main Parts of an AI Agent

An AI agent needs these components to work:



Goal

What the AI is trying to achieve

Reasoning

How the AI thinks about the problem

Planning

Breaking the goal into steps

Memory

Remembering what happened before

Tools

Search, calculator, database

Feedback

Information on whether the plan is working

Where Can We Use Agentic AI?



Software Development

AI writes code and fixes bugs automatically



Customer Support

AI handles customer problems without human help



Research

AI searches and summarizes findings



Personal Assistants

Schedules meetings, sends emails, manages tasks



Business Automation

Handles repetitive office work

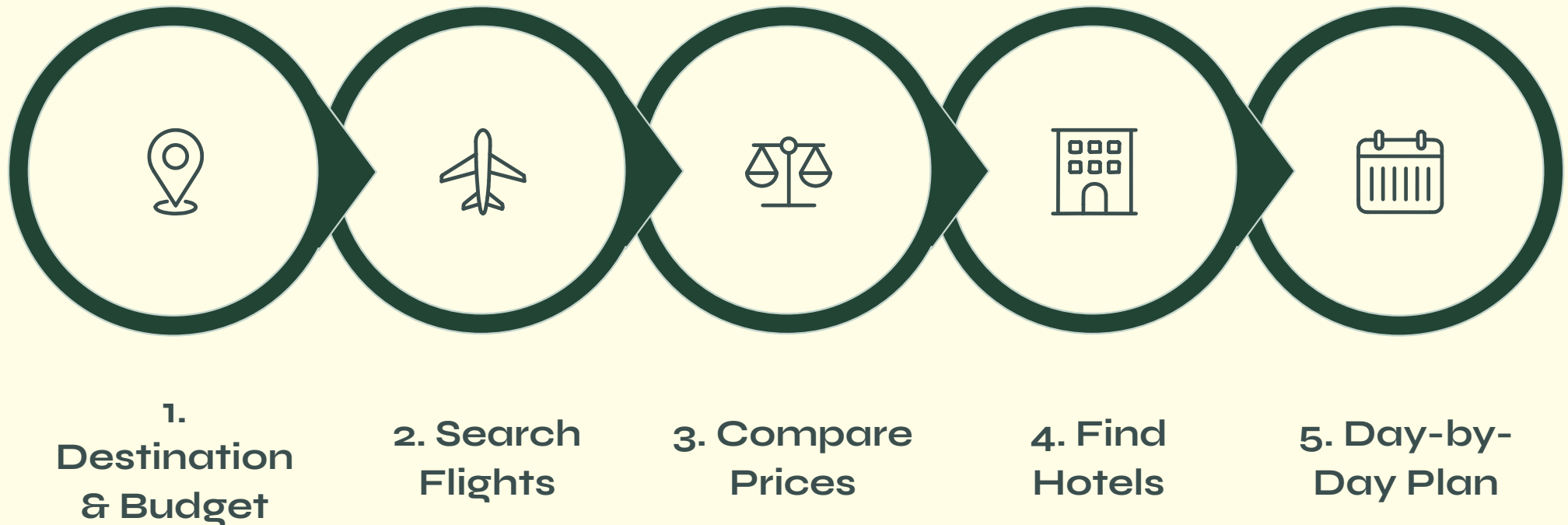


Education

Tutors students and creates personalized learning plans

Example: Planning a Trip

Here's how an AI agent could handle a real task from start to finish:



□ The AI works through every step on its own – but it checks with the user before taking important actions.

Advantages of Agentic AI

Saves time

Automates multi-step tasks that would take humans much longer



Faster Task Completion



No Repetitive Fatigue

Works continuously

Can run 24/7 without getting tired or needing breaks



Checks Its Own Results



Uses Multiple Tools

Handles complexity

Uses different tools and resources to solve hard problems

Reduces manual effort

Takes over repetitive work so people can focus on other things

Problems and Risks

Agentic AI is **not perfect**. We need to be careful:

Wrong Decisions

AI might misunderstand the goal and do the wrong thing

Privacy Issues

AI might access or misuse sensitive data

"Agentic AI is not about generating answers. It is about understanding a goal, planning the steps, and taking action."

Thank You

Security Problems

AI systems could be hacked or manipulated

Costly Mistakes

Errors without human approval can be expensive

Human Supervision

Still very important – AI should not act fully alone

High Costs

Running complex AI systems requires significant resources

Lack of Transparency

We don't always understand why AI makes certain decisions